

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling Forage-Based Livestock Feed Innovations for Resilient Rural Livelihoods in Ethiopia and Zambia

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.3 · Genetics, Inputs & Services for Animals & Aquaculture

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

Ethiopia and Zambia face a critical productivity bottleneck in their livestock sectors: inadequate, expensive, and climate-vulnerable feed systems. In Ethiopia, smallholders rely on declining natural pastures and poor-quality crop residues, while commercial feeds remain unaffordable. In Zambia, high feed costs, climate-induced livestock losses, and limited market access constrain rural incomes. This solution track aims to transform these systems by commercializing and scaling forage-based feed innovations through digitally-enabled bundles, market-led seed systems, and inclusive business models that empower women and youth.

The solution integrates two complementary country-specific bundles. In Ethiopia, a mature, digitally-enabled forage bundle integrates national data pooling, feed balance mapping, and tailored utilization strategies to boost dairy, poultry, and feedlot productivity. In Zambia, the innovation empowers farmers to cultivate and home-mix nutritionally complete ruminant diets using high-protein legumes (mucuna and lablab), proven to compete with expensive commercial feeds. This is not a single technology but a systemic approach combining improved forages, advisory services, seed system development, and SME creation.

Decades of CGIAR research underpin this scaling effort. Pilots under Africa RISING, SAPLING, and AICCRA demonstrate measurable impact: doubled milk yields in Ethiopia, and beef net profits increasing by 7–10% in Zambian trials. With strong demand from national ministries, development partners (World Bank, IFAD, SNV), and private sector associations, the track is poised for responsible scaling. Core implementing partners include ILRI (lead), the Alliance of Bioversity International and CIAT (forage network), and CIMMYT (Zambia), working in close alignment with government extension systems and private enterprises.

2. Innovation Bundle and Scaling Ambition

2.1 What is Being Scaled

- **Ethiopia:** Digitally-enabled forage feed bundle, including feed balance mapping, tailored utilization strategies for dairy/poultry/feedlots, and market-led forage seed systems.
- **Zambia:** Smallholder cultivation of high-protein legumes (mucuna, lablab) for home-mixed ruminant diets; farm-level feed formulation and commercialization.
- **Cross-cutting:** Behavioral change communication, digital extension advisories, business development support for forage seed/feed enterprises, and inclusive business models targeting women and youth.

2.2 Evidence of Impact

- **Ethiopia:** Pilots under Africa RISING and SAPLING validated niche-compatible forage species; adoption driven by bundled advisory services increased farmer incomes (source:).
- **Zambia:** Beef pen-feeding trials increased net profit by 7–10% versus commercial supplements. Mucuna/lablab-based rations raised dairy farmer incomes by 16.5–17%. Pen-fed beef animals doubled market value in 60 days, with average gross income reaching \$650/year (source:).
- **Readiness:** The Ethiopia innovation achieved a scaling readiness score of 6 in 2022; Zambia achieved a score of 8.

2.3 Scaling Targets

Outcome Metric		2026 Target	Long-term Goal (3–5 years)
Smallholder farmers adopting forage-based innovations		53,234 (≥40% women, ≥30% youth) (<i>collective, accountable target</i>)	Ethiopia: 250,000 farmers; Zambia: 500,000 individuals reached
Area under improved forage/ practices		Zambia: 100,000 hectares under leguminous forage cultivation	Expansion based on adoption rates
Jobs supported		Ethiopia: 10,000 farm jobs created/improved; Zambia: 10,000 jobs through new rural feed enterprises	Sustained growth through SME development
Fodder and seed production		Zambia: 10,000 tonnes of fodder and seed produced	Scaled through market systems
Investment leveraged		Zambia: \$1 million through loans, feed sales, bilateral projects	Mobilize AfDB, Gates Foundation, Mastercard Foundation, World Bank/ IFAD programs

3. Scaling Pathway and Theory of Change

3.1 Pathway Strategy

The scaling pathway employs a dual-track approach to transition from donor dependence to market-driven, sustainable systems.

- **Market Systems Development (Private Sector-led):** In both countries, the primary engine for scale is the commercialization of forage seed and feed. This involves strengthening private forage seed enterprises (e.g., Ethiopian Forage Seed Production and Marketing Association), supporting farmer cooperatives to produce and market biomass, and developing rural feed SMEs. S4I funding catalyzes this by providing business development support, market intelligence, and linkages to finance (e.g., INDO Bank in Zambia, microfinance institutions).
- **Public Sector Leverage (Government Integration):** The track systematically integrates with national extension systems to achieve reach at scale. In Zambia, mass communication (radio, TV) will reach 500,000 individuals, and 400 extension personnel will be trained. In Ethiopia, the pathway leverages flagship government programs like the Livestock and Fishery Sector Development Program. This ensures that S4I's technical innovations are embedded within public infrastructure.
- **Coordination as a Catalytic Function:** A core function of S4I funding is to convene the numerous actors investing in forage development. In Ethiopia, ILRI will facilitate multi-stakeholder platforms at national and sub-national levels to co-develop a coherent scaling strategy, ensuring that investments from SNV, USAID, and the government are aligned and mutually reinforcing rather than fragmented.

3.2 Theory of Change

If smallholders, women, and youth gain access to proven, climate-resilient forage technologies and bundled advisory services through functional public and private extension channels, **and if** critical market and coordination failures—such as free seed distribution undermining markets, weak seed systems, and policy barriers—are addressed through multi-stakeholder platforms, policy dialogue, and business

incubation, **then** farmers will adopt improved feed practices, leading to increased milk and meat productivity and household incomes. This will contribute to the **long-term goal** of establishing robust, inclusive, and climate-resilient feed innovation systems that drive rural livelihoods and industrialize the livestock sector across Africa.

3.3 Key Activities Funded by S4I

- **Coordination and Strategy:** Establish and convene national/sub-national multi-stakeholder platforms to map partner activities and co-develop scaling strategies for Ethiopia and Zambia.
- **Market Development:** Provide business development support to farmer cooperatives, youth-led enterprises, and SMEs for forage seed production, biomass marketing, and feed formulation.
- **Capacity Building:** Train extension personnel (400 in Zambia, 1,000 in Ethiopia) on livestock feed and nutrition; deliver behavioral change communication via radio, digital tools, and community engagement.
- **Policy Engagement:** Initiate structured policy dialogue to address bottlenecks (e.g., free seed distribution in Ethiopia, restrictive seed registration policies in Zambia).
- **Innovation Refinement:** Integrate AI-driven advisory tools (with CIAT/ILRI) and validate business models through Living Labs in target regions (Sidama, Ethiopia; Eastern and Southern Provinces, Zambia).

4. Partners and Roles

Partner Type	Specific Partners and Quantities	Role in Scaling Pathway
Demand and Market Partners	Ethiopia: Ministry of Agriculture, Regional Bureaus of Agriculture, Ethiopian Forage Seed Production and Marketing Association (EFSPMA), private forage seed enterprises, farmer cooperatives, SNV, ATI. Zambia: Microfinance institutions (INDO Bank), agricultural retailers (Modern Bazaar), insurance (Nico Insurance), seed houses (K2), butchers/feedlots (Mans Butchery), Zambeef.	Create and sustain demand for forage innovations; provide last-mile distribution of seeds and feeds; finance adoption; aggregate and market livestock products.
Scaling and Technical Partners	Ethiopia: ILRI (lead), Alliance of Bioversity-CIAT (forage network, seed systems), ICARDA, national research systems (NARES), regional bureaus, AICCRA, TAAT. Zambia: CIMMYT, IWMI (water/finance), Alliance of Bioversity-CIAT, NARES (Meskera Research Station), public extension services, EU/SNV/World Bank projects (ESLIP).	Provide technical backstopping, training, and quality assurance; conduct on-farm testing; deliver extension messages; leverage bilateral investments for aligned scaling.

Funding Model: S4I catalytic funding covers coordination, technical assistance, capacity building, and the development of scaling strategies and business models; core inputs (seeds, feed) and large-scale service delivery are financed by partners, government programs, and private sector investment through cost-recovery and commercial partnerships.

5. Enabling Environment and Risks

5.1 Opportunities

- **Strong Government Alignment:** Innovations are integrated into Ethiopia's Food Systems Transformation Pathway and National Dairy Platform, and align with Zambia's government livestock programs.
- **Major Donor Co-investment:** Significant leveraged funding from World Bank, IFAD, AfDB, EU, SNV, and USAID creates a conducive ecosystem for scaling.
- **Private Sector Appetite:** Emergence of forage seed associations (EFSPMA) and interest from agribusinesses (K2 Seed House, INDO Bank) demonstrate growing market demand.
- **Digital Reach Potential:** Mass media and digital extension platforms in both countries offer cost-effective channels to reach hundreds of thousands of farmers.

5.2 Challenges and Mitigation

- **Policy: Indiscriminate free seed distribution (Ethiopia):** Implement structured policy dialogue with the Ministry to develop clear guidelines distinguishing emergency relief from commercial market development.
- **Market: Bulky forage biomass increases transport costs:** Promote cost-effective densification techniques (baling, compacting) to improve logistics for small-scale suppliers.
- **Capacity: Knowledge gaps in livestock extension (Zambia):** Prioritize training-of-trainers for 400 extension personnel and engage private extension providers.
- **Regulatory: Restrictive seed/feed market policies (Zambia):** Collaborate with IWMI and AoW3 to support policy and regulatory easing for rural SME entry.
- **Social: Transitioning from free to paid seed systems:** Engage communities through participatory approaches and targeted campaigns to build trust in commercial sources.

6. Learning and Adaptation

6.1 Monitoring Focus

- **Process Outcomes:** Functionality of multi-stakeholder platforms; number of coordination meetings; partner co-investment leveraged.
- **Behavior Change:** Adoption rates of improved forages and feed rations; shift from free to purchased seed; farmer participation in markets.
- **System Outcomes:** Policy changes (e.g., seed distribution guidelines); emergence of new forage enterprises; jobs created for women and youth.
- **Performance Against Targets:** Quarterly tracking of reach (disaggregated by gender/youth), area under cultivation, and fodder production via digital CRM systems and extension reports.

6.2 Adaptive Management

- **Quarterly Check-ins:** Regular reviews with the S4I PMU to assess progress against milestones and refine activities.
- **Annual Reflection Workshops:** Convene partners and beneficiaries to review learning, share insights from MELIA studies, and co-design adaptations.

- **Real-time Feedback Loops:** Utilize partner reports and stakeholder feedback to identify bottlenecks early and adjust implementation (e.g., refining messaging in behavioral change campaigns).

7. Next Steps and Timeline (2026)

Quarter	Key Activities
Q1	Refine scaling strategies with partners; convene national/sub-national multi-stakeholder platforms in Ethiopia and Zambia; finalize partnership agreements; initiate policy dialogue.
Q2	Roll out capacity building for extension personnel (Zambia); establish Living Labs in Sidama (Ethiopia) and Eastern/Southern Provinces (Zambia); begin business development support for forage seed/feed enterprises.
Q3	Implement behavioral change communication campaigns (radio, digital); facilitate market linkages for producer groups; continue policy engagement; monitor early adoption.
Q4	Conduct mid-term review and impact assessment; document lessons learned; refine Year 2 work plan based on evidence; showcase successful business models.

Communications and Knowledge: Key products will include the co-developed scaling strategy documents, policy briefs, success stories from Living Labs, and digital content (radio spots, mobile advisories) shared via extension networks and partner platforms.

8. Contact

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